2.6 The Binomial Theorem, the Frobenius Map, and Fermat's Little Theorem

In $\mathbb{F}_5$, compute $(1+2)^5$ and 1^5+2^5. The first is $3^5 = 243$, and $243 = 48\cdot 5+3$, so it is 3. The second is $1+32 = 33 = 6\cdot 5+3$, so it is 3 as well. The rule "$(x+y)^n = x^n+y^n$", the classic error of school algebra, is a theorem when n is the characteristic p of the field. This section proves it, and deduces Fermat's little theorem from it with the rigidity of Section 2.5: a homomorphism of $\mathbb{F}_p$ to itself is the identity.

The tool is the binomial theorem. School proves it for numbers; we need it in an arbitrary commutative ring, with the binomial coefficients acting as integer multiples, and so we first set up finite sums in a ring.

2.6.1 Sums and powers in a commutative ring

Sets and Functions, Chapter 3 defines finite sums, products and powers of real numbers by recursion (Sets and Functions, Definition 3.3.1, Sets and Functions, Definition 3.3.2, both from Sets and Functions, Theorem 3.3.1). The definitions use nothing about $\mathbb{R}$ except an addition with a 0 and a multiplication with a 1. So, read on $\mathbb{N} = \omega$ through Number Systems: From \(\nnN\) to \(\nR\), Corollary 1.4.3 (b), with $X = R$, they define in every commutative ring R:

- sums $\sum_{k=m}^{n} a_k$ of elements $a_k \in R$, with the empty sum 0, and the three remarks after Sets and Functions, Definition 3.3.1: a sum depends only on the terms it adds (finite lists), **shifting the index**, and **splitting off the last term**, $\sum_{k=m}^{n+1} a_k = \sum_{k=m}^{n} a_k + a_{n+1}$ for $m \le n+1$;
- powers $x^0 := 1$, $x^{n+1} := x^n x$ for $x \in R$ and $n \in \mathbb{N}$, with the laws of Sets and Functions, Proposition 3.3.4 (a)–(c):

$$x^{m+n} = x^m x^n, \qquad (x^m)^n = x^{mn}, \qquad (xy)^n = x^n y^n.$$

The proofs of those laws in *Sets and Functions* use only (F1)–(F7), (F9) and induction, so they run unchanged in R. We use this paragraph as a template and do not repeat it. (*Analytic Preliminaries*, Chapter 1 proves the same power laws for every commutative ring as Analytic Preliminaries, Proposition 1.1.3; we compare, and cite the template.) In particular $1^n = 1$ for every n, by induction: $1^0 = 1$ and $1^{n+1} = 1^n \cdot 1 = 1$.

The binomial theorem rearranges sums term by term, and the rules it needs are collected in the next lemma. The sums start at $k = 0$; by the definition, $\sum_{k=0}^{0} a_k = 0 + a_0 = a_0$.

Lemma 2.6.1. Finite Sums in a Commutative Ring

Let R be a commutative ring, $n \in \mathbb{N}$, $c \in R$, and let $(a_k)_{k\in\mathbb{N}}$ and $(b_k)_{k\in\mathbb{N}}$ be sequences in R.

(a) $\sum_{k=0}^{n}(a_k + b_k) = \sum_{k=0}^{n} a_k + \sum_{k=0}^{n} b_k$.
(b) $c\sum_{k=0}^{n} a_k = \sum_{k=0}^{n} c\,a_k$.
(c) $\sum_{k=0}^{n} m \cdot a_k = m \cdot \sum_{k=0}^{n} a_k$ for every $m \in \mathbb{Z}$.
(d) Let $j \in \mathbb{N}$, and suppose $a_k = 0$ for every $k \neq j$. Then $\sum_{k=0}^{n} a_k = a_j$ if $j \le n$, and $\sum_{k=0}^{n} a_k = 0$ if $j > n$.
(e) $-\sum_{k=0}^{n} a_k = \sum_{k=0}^{n}(-a_k)$.

(f) *(Splitting off the first term.)* $\sum_{k=0}^{n+1} a_k = a_0 + \sum_{k=0}^{n} a_{k+1}$.

In words: a finite sum may be split termwise, multiplied termwise, and negated termwise; a sum with one non-zero term is that term; and the first term may be split off, the rest being re-indexed from 0. By the finite-lists remark, the lemma applies to finitely many terms $a_0, \dots, a_n$ as well.

Idea. A sum up to $n+1$ is the sum up to n plus one term, so every rule that holds for one term and survives adding one more term holds for all sums: an induction on n whose step is a single axiom, (F1)–(F2) for (a), (F9) for (b), and associativity for (f). Parts (c) and (e) need no induction, because a multiple is a product, $m \cdot a = (m \cdot 1)a$, so they are (b) in disguise.

Proof. Each part is an induction on n (Logic and Proof, Axiom 4.2.1), splitting off the last term at the step, except (c) and (e), which follow from (b).

(a) For $n = 0$ both sides are $a_0 + b_0$. If (a) holds for n, then splitting off the last term,

$$\sum_{k=0}^{n+1}(a_k + b_k) = \Big(\sum_{k=0}^{n} a_k + \sum_{k=0}^{n} b_k\Big) + (a_{n+1} + b_{n+1}),$$

which is $\left(\sum_{k=0}^{n} a_k + a_{n+1}\right) + \left(\sum_{k=0}^{n} b_k + b_{n+1}\right)$ by (F1) and (F2), and that is the right side for $n+1$.

(b) For $n = 0$ both sides are ca_0. At the step, $c\left(\sum_{k=0}^{n} a_k + a_{n+1}\right) = c\sum_{k=0}^{n} a_k + ca_{n+1}$ by (F9), and the induction hypothesis rewrites the first term.

(c) By Lemma 2.5.1 (c), which holds for $m \in \mathbb{Z}$ by part (d) there, $m \cdot a_k = (m \cdot 1)a_k$ and $m \cdot \sum a_k = (m \cdot 1)\sum a_k$. So (c) is (b) with $c = m \cdot 1$.

(d) For $n = 0$: if $j = 0$ the sum is $a_0 = a_j$; if $j > 0$, then $0 \neq j$, so the sum is $a_0 = 0$. Suppose (d) holds for n, and split off a_{n+1}. If $j \leq n$, the sum up to n is a_j and $a_{n+1} = 0$ (as $n+1 \neq j$), so the total is a_j. If $j = n+1$, the sum up to n is 0 (as $j > n$) and the total is $0 + a_{n+1} = a_j$. If $j > n+1$, both parts are 0.

(e) By (c) with $m = -1$, since $(-1) \cdot x = -(1 \cdot x) = -x$ (Lemma 2.5.1 (d)).

(f) For $n = 0$ both sides are $a_0 + a_1$. If (f) holds for n, then

$$\sum_{k=0}^{n+2} a_k = \Big(a_0 + \sum_{k=0}^{n} a_{k+1}\Big) + a_{n+2} = a_0 + \sum_{k=0}^{n+1} a_{k+1},$$

by splitting off the last term, the induction hypothesis, (F2), and splitting off the last term of the sequence $(a_{k+1})_k$, whose term at $k = n+1$ is a_{n+2}. This proves the lemma.

■

2.6.2 Binomial coefficients

In $(x+y)^n$, each product $x^k y^{n-k}$ appears a certain number of times, and that number is the binomial coefficient. We define it by Pascal's rule, which is how the coefficients arise when $(x+y)^n$ is multiplied by $x+y$, and we never divide.

Definition 2.6.1. Binomial Coefficients

The **binomial coefficients** $\binom{n}{k} \in \mathbb{N}$, for $n, k \in \mathbb{N}$, are defined by recursion on n (Sets and Functions, Theorem 3.3.1, with X the set of functions $\mathbb{N} \to \mathbb{N}$):

$$\binom{0}{0} := 1, \qquad \binom{0}{k} := 0 \quad (k \geq 1),$$
$$\binom{n+1}{0} := 1, \quad \binom{n+1}{k+1} := \binom{n}{k} + \binom{n}{k+1} \quad (k \in \mathbb{N}).$$

In words: row $n+1$ starts with 1, and each later entry is the sum of the two entries of row n above it (**Pascal's rule**). Each row is a whole function $\mathbb{N} \to \mathbb{N}$, so the recursion has values for every k, and a coefficient is a natural number, never a quotient.

Three facts follow by one induction on n each.

- $\binom{n}{k} = 0$ for $k > n$. For $n = 0$ this is the definition. If it holds for n and $k > n+1$, write $k = j+1$ with $j > n$; then $\binom{n+1}{j+1} = \binom{n}{j} + \binom{n}{j+1} = 0 + 0$.
- $\binom{n}{n} = 1$. Indeed $\binom{0}{0} = 1$, and $\binom{n+1}{n+1} = \binom{n}{n} + \binom{n}{n+1} = 1 + 0$ by the first fact.
- $\binom{n}{1} = n$. Indeed $\binom{0}{1} = 0$, and $\binom{n+1}{1} = \binom{n}{0} + \binom{n}{1} = 1 + n$.

The rows $n = 0, \ldots, 6$, with the zeros for $k > n$ shown:

n	$k=0$	1	2	3	4	5	6
0	1	0	0	0	0	0	0
1	1	1	0	0	0	0	0
2	1	2	1	0	0	0	0
3	1	3	3	1	0	0	0
4	1	4	6	4	1	0	0
5	1	5	10	10	5	1	0
6	1	6	15	20	15	6	1

Remark. The formula $\binom{n}{k} \cdot k! \cdot (n-k)! = n!$ for $k \leq n$ is true, but we neither prove nor use it. The route below needs one identity between neighboring coefficients and avoids division altogether, which matters because in $\mathbb{F}_p$ the factorial $p!$ is 0.

The identity we need relates row $n+1$ to row n by a factor. It is what makes a prime divide the middle coefficients of row p.

Proposition 2.6.2. Absorption Identity

For all $n, k \in \mathbb{N}$,

$$(k+1)\binom{n+1}{k+1} = (n+1)\binom{n}{k}.$$

Idea. The definition is a recursion on n, so we induct on n, proving the statement for **all** k at once; the step then needs the hypothesis at two values of k. In the step, expand the left side by Pascal's rule, split the factor $k+2$ as $(k+1)+1$ in front of one of the two terms, apply the hypothesis twice, and recombine with Pascal's rule. For $n=2$, $k=0$ it reads $1\cdot 3 = 3\cdot 1$; for $n=3$, $k=1$, $2\cdot 6 = 4\cdot 3$.

Proof. We prove by induction on n (Logic and Proof, Axiom 4.2.1) the statement "for every $k\in\mathbb{N}$, $(k+1)\binom{n+1}{k+1} = (n+1)\binom{n}{k}$".

Base $n=0$. For $k=0$: $1\cdot\binom{1}{1} = 1 = 1\cdot\binom{0}{0}$. For $k\geq 1$: $\binom{1}{k+1} = 0$ since $k+1>1$, and $\binom{0}{k}=0$; both sides are 0.

Step. Suppose the statement holds for n. For $k=0$, the claim for $n+1$ reads $\binom{n+2}{1} = (n+2)\binom{n+1}{0}$, and both sides are $n+2$ by the facts above. For $k=j+1$ with $j\in\mathbb{N}$, Pascal's rule gives

$$\begin{aligned}(j+2)\binom{n+2}{j+2} &= (j+2)\binom{n+1}{j+1} + (j+2)\binom{n+1}{j+2}\\ &= (j+1)\binom{n+1}{j+1} + \binom{n+1}{j+1} + (j+2)\binom{n+1}{j+2}\\ &= (n+1)\binom{n}{j} + \binom{n+1}{j+1} + (n+1)\binom{n}{j+1}\\ &= (n+1)\binom{n+1}{j+1} + \binom{n+1}{j+1} = (n+2)\binom{n+1}{j+1},\end{aligned}$$

where the third line uses the hypothesis at $k=j$ and at $k=j+1$, and the fourth line Pascal's rule again. This is the claim for $n+1$ at $k=j+1$. This proves the proposition.

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2.6.3 The binomial theorem

Expanding $(x+y)^n$ multiplies out n factors $x+y$ and collects the products $x^k y^{n-k}$. In a commutative ring this collecting is legitimate, and the counts are the binomial coefficients, acting as integer multiples.

Theorem 2.6.3. The Binomial Theorem in a Commutative Ring

Let R be a commutative ring, $x, y\in R$ and $n\in\mathbb{N}$. Then

$$(x+y)^n = \sum_{k=0}^{n}\binom{n}{k}\cdot\left(x^k y^{n-k}\right).$$

Here $\binom{n}{k}\cdot z$ is the integer multiple of Section 2.5, and $n-k\in\mathbb{N}$ since $k\leq n$.

Idea. Induct on n, with $(x+y)^{n+1} = (x+y)^n x + (x+y)^n y$. Multiplying the sum for $(x+y)^n$ by x raises every power of x by one, which shifts the coefficients one place to the right; multiplying by y leaves them in place. Adding the two rows entry by entry is exactly Pascal's rule. The work is in lining the two sums up over the same range $k = 0, \dots, n+1$: the shifted sum gets a zero first term, and the other one a zero last term.

Proof. Induction on n (Logic and Proof, Axiom 4.2.1). For $n = 0$, $(x+y)^0 = 1$ and the right side is $\binom{0}{0} \cdot (x^0 y^0) = 1 \cdot (1 \cdot 1) = 1$.

Suppose the formula holds for n, and put $t_k \coloneqq \binom{n}{k} \cdot (x^k y^{n-k})$ for $0 \le k \le n$. By the power recursion, (F9) and Lemma 2.6.1 (b) with (F6),

$$(x+y)^{n+1} = (x+y)^n x + (x+y)^n y = \sum_{k=0}^{n} t_k x + \sum_{k=0}^{n} t_k y.$$

By the rule $(m \cdot a)b = m \cdot (ab)$ of Lemma 2.5.1 (d), with (F5), (F6) and the power recursion,

$$t_k x = \binom{n}{k} \cdot (x^{k+1} y^{n-k}), \qquad t_k y = \binom{n}{k} \cdot (x^k y^{n+1-k}).$$

For $0 \le k \le n+1$ let $z_k \coloneqq x^k y^{n+1-k}$, and define

$$u_0 \coloneqq 0, \quad u_k \coloneqq \binom{n}{k-1} \cdot z_k \ (1 \le k \le n+1), \qquad w_k \coloneqq \binom{n}{k} \cdot z_k \ (0 \le k \le n+1).$$

Then $t_k x = u_{k+1}$ and $t_k y = w_k$ for $0 \le k \le n$, and $w_{n+1} = 0 \cdot z_{n+1} = 0$, since $\binom{n}{n+1} = 0$. By Lemma 2.6.1 (f) and $u_0 = 0$, and by splitting off the last term $w_{n+1} = 0$,

$$\sum_{k=0}^{n} t_k x = \sum_{k=0}^{n+1} u_k, \qquad \sum_{k=0}^{n} t_k y = \sum_{k=0}^{n+1} w_k.$$

So $(x+y)^{n+1} = \sum_{k=0}^{n+1} (u_k + w_k)$ by Lemma 2.6.1 (a). Finally, $u_0 + w_0 = 0 + 1 \cdot z_0 = \binom{n+1}{0} \cdot z_0$, and for $k = j+1$ with $0 \le j \le n$, the sum rule of Lemma 2.5.1 (d) and Pascal's rule give

$$u_k + w_k = \binom{n}{j} \cdot z_k + \binom{n}{j+1} \cdot z_k = \binom{n+1}{j+1} \cdot z_k = \binom{n+1}{k} \cdot z_k.$$

Hence $(x+y)^{n+1} = \sum_{k=0}^{n+1} \binom{n+1}{k} \cdot (x^k y^{n+1-k})$, which is the formula for $n+1$. This proves the theorem.

■

The proof used (F6) at every regrouping of $x^k y^{n-k} x$, and the theorem needs it: the linear-algebra volumes meet matrices, whose multiplication is not commutative, and there the expansion holds only for pairs that commute. For $R = \mathbb{Z}$ the theorem is the familiar expansion, since there $m \cdot z = mz$ (Section 2.5).

In row 5 of the table, the four middle entries $5, 10, 10, 5$ are all multiples of 5; in row 6 the middle entries 15 and 20 are not multiples of 6. Primes are special.

Lemma 2.6.4. A Prime Divides the Middle Binomial Coefficients

Let p be a prime and $k \in \mathbb{N}$ with $0 < k < p$. Then $p \mid \binom{p}{k}$.

Proof. We read the coefficients in $\mathbb{N} \subseteq \mathbb{Z}$. By Proposition 2.6.2 with $n = p - 1$ and $k - 1$ in place of k,

$$k\binom{p}{k} = p\binom{p-1}{k-1},$$

so $p \mid k\binom{p}{k}$. Since $0 < k < p$, $p \nmid k$: a positive multiple of p is at least p (Number Systems: From \(\nN\) to \(\nR\), Proposition 2.5.1 (e)). Hence $\gcd(p, k) = 1$ by Corollary 1.2.2 (a), and Theorem 1.2.1 gives $p \mid \binom{p}{k}$. This proves the lemma.

■

The hypothesis "prime" is needed: $\binom{4}{2} = 6$, and $4 \nmid 6$. In the proof, $4 \mid 2\binom{4}{2}$, but $\gcd(4, 2) = 2$, so Euclid's lemma does not apply.

Quick check. Use Proposition 2.6.2 to compute $\binom{7}{3}$ from $\binom{6}{2} = 15$, and check that 7 divides it.

Solution. With $n = 6$, $k = 2$: $3\binom{7}{3} = 7\binom{6}{2} = 7 \cdot 15 = 105$, so $\binom{7}{3} = 35$, and $35 = 5 \cdot 7$. (By Pascal's rule, $\binom{7}{3} = \binom{6}{2} + \binom{6}{3} = 15 + 20 = 35$, the same.)

2.6.4 The Frobenius map

We can now expand $(x + y)^p$ in a field of characteristic p.

Theorem 2.6.5. The Frobenius Homomorphism

Let F be a field of characteristic $p > 0$.

(a) For all $x, y \in F$,

$$(x + y)^p = x^p + y^p, \qquad (xy)^p = x^p y^p,$$

and $1^p = 1$. So the **Frobenius map** $\mathrm{Fr}\colon F \to F$, $\mathrm{Fr}(x) := x^p$, is a homomorphism, and it is injective.

(b) $(x - y)^p = x^p - y^p$ for all $x, y \in F$.

(c) If F is finite, then Fr is an automorphism of F.

By Theorem 2.5.3, the hypothesis $\operatorname{char} F = p > 0$ already makes p prime.

Try it first (ten minutes): expand $(x + y)^5$ and $(x + y)^6$ in $\mathbb{F}_5$, using the rows $n = 5$ and $n = 6$ of the table above, and see which coefficients survive.

Idea. For $(x+y)^5$ the coefficients $1,5,10,10,5,1$ become $1,0,0,0,0,1$ in $\mathbb{F}_5$, and $(x+y)^5 = x^5+y^5$. For $(x+y)^6$ the coefficients $1,6,15,20,15,6,1$ become $1,1,0,0,0,1,1$, and $(x+y)^6 = x^6+x^5y+xy^5+y^6$: the extra terms survive because 6 is not the characteristic. So the plan is: expand by the binomial theorem; every middle coefficient $\binom{p}{k}$ is a multiple of p; and a multiple of p acts on F as 0, because $p \cdot 1 = 0$. The product rule and $1^p = 1$ are power laws, and injectivity is free for a homomorphism of fields.

Before reading on: where is it used that p is prime?

Proof.

(a) By Theorem 2.6.3, $(x+y)^p = \sum_{k=0}^{p} t_k$ with $t_k := \binom{p}{k} \cdot (x^k y^{p-k})$. Let $0 < k < p$. By Lemma 2.6.4, $p \mid \binom{p}{k}$, so $\binom{p}{k} \cdot 1 = 0$ by Proposition 2.5.2 (a), and then, by Lemma 2.5.1 (c) and Theorem 2.1.1 (d),

$$t_k = \left(\binom{p}{k} \cdot 1\right)(x^k y^{p-k}) = 0 \cdot (x^k y^{p-k}) = 0.$$

The end terms are $t_0 = 1 \cdot (x^0 y^p) = y^p$ and $t_p = 1 \cdot (x^p y^0) = x^p$. Let $b_k := t_k$ for $k = 0$ and $b_k := 0$ otherwise, and $c_k := t_k$ for $k = p$ and $c_k := 0$ otherwise. Since $p \neq 0$, $t_k = b_k + c_k$ for every $0 \leq k \leq p$, so by Lemma 2.6.1 (a) and (d),

$$(x+y)^p = \sum_{k=0}^{p} b_k + \sum_{k=0}^{p} c_k = t_0 + t_p = y^p + x^p,$$

which is $x^p + y^p$ by (F1). Next, $(xy)^p = x^p y^p$ is the third power law of this section's opening paragraph (Sets and Functions, Proposition 3.3.4 (c), read in F), and $1^p = 1$. So Fr is a homomorphism (Definition 2.4.1), and it is injective by Proposition 2.4.1 (c).

(b) By Proposition 2.4.1 (a), $\mathrm{Fr}(x-y) = \mathrm{Fr}(x) - \mathrm{Fr}(y)$.

(c) The image $\mathrm{Fr}(F)$ is a subset of F, and Fr, with codomain $\mathrm{Fr}(F)$, is a bijection $F \to \mathrm{Fr}(F)$ by (a). If $\mathrm{Fr}(F) \neq F$, this would be a bijection from the finite set F onto a proper subset, which Sets and Functions, Theorem 4.1.9 forbids. So $\mathrm{Fr}(F) = F$, Fr is bijective, and a bijective homomorphism $F \to F$ is an automorphism (Definition 2.4.1). This proves the theorem.

■

Answer to the prompt. Primality is used in one place: Lemma 2.6.4, whose proof needs Euclid's lemma, and so needs p prime. It costs nothing here, since a positive characteristic is always prime (Theorem 2.5.3).

In words: in characteristic p, raising to the p-th power respects sums. The map is not the identity in general; Exercise C3 of Section 2.5 found that on the four-element field it exchanges two elements. On $\mathbb{F}_p$ itself it is the identity, and that is Fermat's little theorem.

Warning. "$(x+y)^n = x^n + y^n$" is not a general rule. In a field of characteristic $p > 0$ it holds for $n = p$ (Theorem 2.6.5 (a)), and so for $n = p^2$ by Exercise B2 (and in the same way for every power of p). With other exponents it can fail: in $\mathbb{F}_3$, $(1+1)^2 = 4 = 1$, while $1^2+1^2 = 2$. In a ring of composite characteristic it can fail even with n equal to the characteristic: in

$\mathbb{Z}/4\mathbb{Z}$, which has characteristic 4, $([1]+[1])^4 = [16] = [0]$, while $[1]^4 + [1]^4 = [2]$; the middle coefficient $\binom{4}{2} = 6$ is not a multiple of 4. In $\mathbb{Z}$ the identity fails for every $n \geq 2$: $(1+1)^n = 2^n \neq 2$.

2.6.5 Fermat's little theorem

On $\mathbb{F}_p$, the Frobenius map is a homomorphism $\mathbb{F}_p \to \mathbb{F}_p$, and by Corollary 2.5.6 (b) there is only one such map, the identity. That is the whole proof of the first part below.

Theorem 2.6.6. Fermat's Little Theorem

Let p be a prime.

(a) $a^p = a$ for every $a \in \mathbb{F}_p$.
(b) $a^{p-1} = 1$ for every non-zero $a \in \mathbb{F}_p$; so $a^{-1} = a^{p-2}$.
(c) For every $a \in \mathbb{Z}$, $p \mid a^p - a$; and if $p \nmid a$, then $p \mid a^{p-1} - 1$.

Proof.

(a) $\operatorname{char} \mathbb{F}_p = p$ (Example (The characteristics of the standard fields)), so $\mathrm{Fr}\colon \mathbb{F}_p \to \mathbb{F}_p$ is a homomorphism by Theorem 2.6.5 (a). By Corollary 2.5.6 (b), the only homomorphism $\mathbb{F}_p \to \mathbb{F}_p$ is the identity. Hence $a^p = \mathrm{Fr}(a) = a$ for every $a \in \mathbb{F}_p$.

(b) Let $a \neq 0$. Since $p \geq 2$, $p - 1 \in \mathbb{N}$ and $a \cdot a^{p-1} = a^{p-1}a = a^p = a = a \cdot 1$, by (F6), the power recursion and (a). Canceling $a \neq 0$ (Theorem 2.1.1 (b)) gives $a^{p-1} = 1$. Likewise $p - 2 \in \mathbb{N}$ and $a \cdot a^{p-2} = a^{p-1} = 1$, so $a^{-1} = a^{p-2}$ by uniqueness of inverses (Theorem 2.1.1 (c)).

(c) Write $[a]$ for the class of $a \in \mathbb{Z}$ in $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$. Since $[a][b] = [ab]$ (Theorem 1.3.1 (b)), induction on n gives $[a]^n = [a^n]$ for every $n \in \mathbb{N}$: $[a]^0 = [1] = [a^0]$, and $[a]^{n+1} = [a]^n[a] = [a^n][a] = [a^{n+1}]$. By (a), $[a^p] = [a]^p = [a]$, so $p \mid a^p - a$ (Theorem 1.3.1 (a), for the ideal $p\mathbb{Z}$). If $p \nmid a$, then $[a] \neq [0]$ (Theorem 1.3.1 (d)), so $[a^{p-1}] = [a]^{p-1} = [1]$ by (b), and $p \mid a^{p-1} - 1$. This proves the theorem. ∎

Primality is needed in (c). For the modulus 4 and $a = 2$, $2^4 - 2 = 14$ is not a multiple of 4 (in $\mathbb{Z}/4\mathbb{Z}$ this is $([1]+[1])^4 = [0] \neq [2]$, from the warning above), and although $4 \nmid 2$, $2^3 - 1 = 7$ is not a multiple of 4 either.

Part (b) gives a second way to invert in $\mathbb{F}_p$, besides the Euclidean algorithm of Chapter 1: $a^{-1} = a^{p-2}$. It also reduces large exponents: for non-zero a, only the remainder of the exponent on division by $p-1$ matters, since $a^{q(p-1)+r} = (a^{p-1})^q a^r = a^r$. Exercise C2 below gives a second proof of (a), by induction, and Chapter 4 gives another proof of (b), from a product of all the non-zero elements.

Example. Powers and inverses in prime fields

(i) Compute 3^{100} in $\mathbb{F}_7$.
(ii) Compute 2^{100} in $\mathbb{F}_{101}$.

(iii) Find the inverse of 7 in $\mathbb{F}_{31}$ twice: by Theorem 2.6.6 (b), and by the Euclidean algorithm.

(iv) What is the Frobenius map of $\mathbb{F}_p$?

Solution.

(i) 7 is prime (the small-primes remark of Section 1.2) and $3 \neq 0$ in $\mathbb{F}_7$, so $3^6 = 1$ by Theorem 2.6.6 (b). Since $100 = 16 \cdot 6 + 4$, the power laws give $3^{100} = (3^6)^{16}\, 3^4 = 1 \cdot 81$, and $81 = 11 \cdot 7 + 4$. So $3^{100} = 4$ in $\mathbb{F}_7$.

(ii) 101 is prime: by the remark after Lemma 1.2.4 it suffices that no d with $2 \leq d$ and $d \cdot d \leq 101$, that is $2 \leq d \leq 10$, divides 101, and the remainders of 101 on division by $2, 3, \ldots, 10$ are $1, 2, 1, 1, 5, 3, 5, 2, 1$. Since $2 \neq 0$ in $\mathbb{F}_{101}$, Theorem 2.6.6 (b) gives $2^{100} = 2^{101-1} = 1$. No multiplication was needed.

(iii) 31 is prime by the same test ($d = 2, 3, 4, 5$ leave remainders $1, 1, 3, 1$). *By Fermat:* since $7 \neq 0$ in $\mathbb{F}_{31}$, Theorem 2.6.6 (b) gives $7^{-1} = 7^{29}$. Square repeatedly in $\mathbb{F}_{31}$: $7^2 = 49 = 18$, $7^4 = 18^2 = 324 = 14$, $7^8 = 14^2 = 196 = 10$, $7^{16} = 10^2 = 100 = 7$. With $29 = 16 + 8 + 4 + 1$,

$$7^{29} = 7^{16}\, 7^8\, 7^4\, 7 = 7 \cdot 10 \cdot 14 \cdot 7 = 70 \cdot 98 = 8 \cdot 5 = 40 = 9.$$

By Euclid: $31 = 4 \cdot 7 + 3$ and $7 = 2 \cdot 3 + 1$, so

$$1 = 7 - 2 \cdot 3 = 7 - 2(31 - 4 \cdot 7) = 9 \cdot 7 - 2 \cdot 31,$$

and $7^{-1} = 9$ by Theorem 1.4.1. Both routes give 9. Check: $7 \cdot 9 = 63 = 2 \cdot 31 + 1$. For small p the Euclidean algorithm is shorter; Fermat's route needs no back-substitution, only squaring.

(iv) The identity, by Theorem 2.6.6 (a): $\mathrm{Fr}(a) = a^p = a$ for every $a \in \mathbb{F}_p$.

Quick check. Compute 2^6 in $\mathbb{F}_7$ and 3^{10} in $\mathbb{F}_{11}$, with and without Theorem 2.6.6.

Solution. Both are 1. Without it: $2^6 = 64 = 9 \cdot 7 + 1$; and $3^5 = 243 = 22 \cdot 11 + 1$, so $3^{10} = (3^5)^2 = 1$. With it: 7 and 11 are prime, $2 \neq 0$ in $\mathbb{F}_7$ and $3 \neq 0$ in $\mathbb{F}_{11}$, and the exponents are $7 - 1$ and $11 - 1$.

2.6.6 Exercises

2.6.6.1 A. Check your understanding

Exercise. A1

These parts are about Theorem 2.6.5 and its proof.

(a) Where in the proof of (a) is it used that p is prime, and why does the theorem not need "p prime" as a separate hypothesis?

(b) List $\binom{5}{k}$ for $0 \le k \le 5$ and $\binom{6}{k}$ for $0 \le k \le 6$. Which are divisible by 5, and which by 6?

(c) Why is Fr injective? Does the proof use that F is finite?

(d) State the idea of the proof of (a) in two sentences.

Solution.

(a) In Lemma 2.6.4, through Euclid's lemma: $p \mid k\binom{p}{k}$ gives $p \mid \binom{p}{k}$ only because $\gcd(p,k) = 1$, which needs p prime. A positive characteristic is always prime (Theorem 2.5.3), so "char $F = p > 0$" already includes it.

(b) Row 5: $1, 5, 10, 10, 5, 1$; the four middle ones are divisible by 5, the end ones are not. Row 6: $1, 6, 15, 20, 15, 6, 1$; only the two 6's are divisible by 6, and 15, 20 are not. This is why $(x+y)^6 \neq x^6 + y^6$ in a ring of characteristic 6 such as $\mathbb{Z}/6\mathbb{Z}$.

(c) Every homomorphism of fields is injective (Proposition 2.4.1 (c)): $\mathrm{Fr}(a) = 0$ only for $a = 0$. The proof of injectivity does not use finiteness; finiteness is used only for surjectivity in (c).

(d) Expand $(x+y)^p$ by the binomial theorem; every middle coefficient $\binom{p}{k}$ is a multiple of p. In characteristic p, $p \cdot 1 = 0$, so every middle term vanishes and only $x^p + y^p$ remains.

2.6.6.2 B. Practice

Exercise. B1

Compute 5^{38} in $\mathbb{F}_{13}$, and find the inverse of 5 in $\mathbb{F}_{13}$ by Theorem 2.6.6.

Solution. 13 is prime (no d with $2 \le d \le 3$ divides it, and $4 \cdot 4 > 13$; the remark after Lemma 1.2.4), and $5 \neq 0$ in $\mathbb{F}_{13}$. By Theorem 2.6.6 (b), $5^{12} = 1$. Since $38 = 3 \cdot 12 + 2$, $5^{38} = (5^{12})^3\, 5^2 = 25 = 12$ in $\mathbb{F}_{13}$. For the inverse, $5^{-1} = 5^{11}$. Now $5^2 = 12 = -1$, so $5^{10} = (5^2)^5 = (-1)^5 = -1$ and $5^{11} = -5 = 8$. Check: $5 \cdot 8 = 40 = 3 \cdot 13 + 1$. Hence $5^{38} = 12$ and $5^{-1} = 8$.

Exercise. B2

Let F be a field of characteristic $p > 0$.

(a) Show that $(x+y)^{p^2} = x^{p^2} + y^{p^2}$ for all $x, y \in F$.

(b) Show that $\left(\sum_{i=1}^{m} x_i\right)^p = \sum_{i=1}^{m} x_i^p$ for all $m \in \mathbb{N}$ and $x_1, \ldots, x_m \in F$.

Solution.

(a) $\mathrm{Fr} \circ \mathrm{Fr}$ is a homomorphism, as a composite of homomorphisms (Proposition 2.4.1 (e)), and $(\mathrm{Fr} \circ \mathrm{Fr})(x) = (x^p)^p = x^{p^2}$ by the power law $(x^m)^n = x^{mn}$. Applying its additivity to $x + y$ gives the claim.

(b) Induction on m. For $m = 0$ both sides are 0: the empty sum is 0, and $0^p = 0^{p-1} \cdot 0 = 0$

as $p \geq 1$. If the claim holds for m, split off the last term and use Theorem 2.6.5 (a):

$$\Big(\sum_{i=1}^{m+1} x_i\Big)^p = \Big(\sum_{i=1}^{m} x_i + x_{m+1}\Big)^p = \Big(\sum_{i=1}^{m} x_i\Big)^p + x_{m+1}^p = \sum_{i=1}^{m+1} x_i^p.$$

Exercise. B3

Let p be a prime. Show that the solutions $x \in \mathbb{F}_p$ of $x^{p-1} = 1$ are exactly the non-zero elements, and that every $a \in \mathbb{F}_p$ satisfies $a^p - a = 0$.

Solution. Every non-zero a satisfies $a^{p-1} = 1$ by Theorem 2.6.6 (b). The element 0 does not: since $p - 1 \geq 1$, $0^{p-1} = 0^{p-2} \cdot 0 = 0 \neq 1$. So the solutions are exactly the non-zero elements. And $a^p - a = a - a = 0$ for every a by Theorem 2.6.6 (a).

2.6.6.3 C. Going deeper

Exercise. C1: Exit check: the Frobenius identity

Let F be a field of characteristic $p > 0$. Using Theorem 2.6.3 and Lemma 2.6.4, prove that $\mathrm{Fr}\colon F \to F$, $x \mapsto x^p$, is an injective homomorphism.

Solution. Sums. By Theorem 2.6.3, $(x+y)^p = \sum_{k=0}^{p} \binom{p}{k} \cdot (x^k y^{p-k})$. For $0 < k < p$, $\binom{p}{k} \cdot (x^k y^{p-k}) = (\binom{p}{k} \cdot 1)(x^k y^{p-k})$ by Lemma 2.5.1 (c). By Lemma 2.6.4, $p \mid \binom{p}{k}$, and since $\operatorname{char} F = p$, $\binom{p}{k} \cdot 1 = 0$ (Proposition 2.5.2 (a)); so the term is 0 (Theorem 2.1.1 (d)). The terms $k = 0$ and $k = p$ are $1 \cdot y^p = y^p$ and $1 \cdot x^p = x^p$, since $\binom{p}{0} = \binom{p}{p} = 1$ and $x^0 = y^0 = 1$. A sum whose terms vanish except at two places is the sum of those two terms (Lemma 2.6.1 (a), (d)), so $(x+y)^p = x^p + y^p$.

Products and one. $(xy)^p = x^p y^p$ by the power law $(xy)^n = x^n y^n$, and $1^p = 1$. So Fr is a homomorphism (Definition 2.4.1).

Injective. Every homomorphism of fields is injective (Proposition 2.4.1 (c)).

Marking scheme. (10 points)

- 2: the binomial expansion written with integer multiples $\binom{p}{k} \cdot (x^k y^{p-k})$.
- 2: each middle term reduced to a multiple that vanishes, with the rule named (for example $(\binom{p}{k} \cdot 1)(x^k y^{p-k})$ by $(m \cdot 1)a = m \cdot a$, or $\binom{p}{k} \cdot z = q \cdot (p \cdot z)$ where $\binom{p}{k} = qp$).
- 2: that multiple is 0 from $p \mid \binom{p}{k}$ and $\operatorname{char} F = p$, with both reasons named.
- 1: the end terms y^p and x^p.
- 1: $(xy)^p = x^p y^p$ and $1^p = 1$.
- 2: injectivity, by Proposition 2.4.1 (c) or by any correct direct argument.

Second route for injectivity: $x^p = y^p \Rightarrow (x-y)^p = 0 \Rightarrow x = y$ earns the 2 points, 1 for $(x-y)^p = x^p - y^p$ with its reason (Proposition 2.4.1 (a), or $(-y)^p = -y^p$ for every prime p, the even one included), and 1 for $x - y = 0$ because a field has no zero divisors.

Fatal errors (0 for the step): "$\binom{p}{k} = 0$" as an integer; using $(x+y)^p = x^p + y^p$ in its own proof.

Exercise. C2: Fermat by induction

Let p be a prime. Give another proof of Theorem 2.6.6 (a) and of the first claim of Theorem 2.6.6 (c), without Corollary 2.5.6. In (a) use only Theorem 2.6.5 (a) (with $\operatorname{char} \mathbb{F}_p = p$, Example (The characteristics of the standard fields)) and Theorem 1.3.3 (b), (d); in (b) you may also use Theorem 1.3.1.

(a) prove by induction on $r \in \mathbb{N}$ that $(r \cdot 1)^p = r \cdot 1$ in $\mathbb{F}_p$, and deduce $a^p = a$ for every $a \in \mathbb{F}_p$;

(b) deduce that $p \mid a^p - a$ for every integer a.

Solution.

(a) For $r = 0$: $(0 \cdot 1)^p = 0^p = 0^{p-1} \cdot 0 = 0 = 0 \cdot 1$, as $p \geq 1$. If $(r \cdot 1)^p = r \cdot 1$, then by Theorem 2.6.5 (a) ($\operatorname{char} \mathbb{F}_p = p$, Example (The characteristics of the standard fields)),

$$((r+1) \cdot 1)^p = (r \cdot 1 + 1)^p = (r \cdot 1)^p + 1^p = r \cdot 1 + 1 = (r+1) \cdot 1.$$

Every element of $\mathbb{F}_p$ is $[r]$ for some $r \in \{0, \ldots, p-1\}$ (Theorem 1.3.3 (b)), and $[r] = r \cdot [1]$ (part (d) there). So $a^p = a$ for every $a \in \mathbb{F}_p$.

(b) Let $a \in \mathbb{Z}$. Then $[a] \in \mathbb{F}_p$, so $[a]^p = [a]$ by (a). By induction on n, $[a]^n = [a^n]$, using $[a][b] = [ab]$ (Theorem 1.3.1 (b)), as in the proof of Theorem 2.6.6 (c). So $[a^p] = [a]$, and $p \mid a^p - a$ (Theorem 1.3.1 (a), for the ideal $p\mathbb{Z}$). (This proves only what Theorem 2.6.6 already gives; it is practice with the Frobenius identity, and nothing cites it.)

Exercise. C3: Roots of order p

Let F be a finite field of characteristic p. Show that for every $a \in F$ the equation $x^p = a$ has **exactly one** solution $x \in F$. Deduce that in $\mathbb{F}_p$, the only solution of $x^p = a$ is $x = a$.

Solution. By Corollary 2.5.7 (a), p is a prime, so $p > 0$, and by Theorem 2.6.5 (c), Fr is a bijection $F \to F$. Surjectivity gives a solution of $\operatorname{Fr}(x) = a$, and injectivity says there is at most one. In $\mathbb{F}_p$, $x = a$ is a solution by Theorem 2.6.6 (a), $a^p = a$, so it is the only one. The uniqueness half needs only that Fr is injective, so it holds in every field of characteristic $p > 0$ (Theorem 2.6.5 (a)); existence used the finiteness of F.

2.6.6.4 R. Review

Exercise. R1

Find all $z \in \mathbb{C}$ with $z^2 = -3 + 4i$, and explain why there are exactly two.

Solution. By the square-root formula of Analytic Preliminaries, Proposition 1.4.1, with $a = -3$, $b = 4$: $r = |-3 + 4i| = 5$ (since $9 + 16 = 25$), $(r + a)/2 = 1$, $(r - a)/2 = 4$, and $b > 0$, so $w = 1 + 2i$. Check: $(1 + 2i)^2 = 1 + 4i + 4i^2 = -3 + 4i$. By Corollary 2.1.2 in the field $\mathbb{C}$ (Example (The Fields $\mathbb{Q}$, $\mathbb{R}$ and $\mathbb{C}$)), $z^2 = w^2$ if and only if $z = w$ or $z = -w$. So the solutions are $1 + 2i$ and $-1 - 2i$. They are two, not one: $w = -w$ would give $2w = 0$, hence $w = 0$, since $2 \neq 0$ in $\mathbb{C}$ ($\operatorname{char} \mathbb{C} = 0$, Example (The characteristics of the standard fields), with Proposition 2.5.2 (b)).

Exercise. R2

Show that 7 has no square root in $\mathbb{Q}(\sqrt{2})$.

Solution. Suppose $(a+b\sqrt{2})^2 = 7$ with $a, b \in \mathbb{Q}$. Expanding, $(a^2+2b^2)+2ab\sqrt{2} = 7+0\sqrt{2}$, so by unique representation (@prp-cof-sqrt2-field-archimedean (a)), $2ab = 0$ and $a^2 + 2b^2 = 7$. Since $2 \neq 0$ in $\mathbb{Q}$ ($\operatorname{char} \mathbb{Q} = 0$, Example (The characteristics of the standard fields), with Proposition 2.5.2 (b)), Theorem 2.1.1 (f) applied twice (to $2 \cdot (ab) = 0$, then to $ab = 0$) gives $a = 0$ or $b = 0$. If $b = 0$, then $a^2 = 7$. If $a = 0$, then $2b^2 = 7$, so $(2b)^2 = 14$. Both are excluded by the claim stated with @prp-cof-sqrt2-field-incomplete: no natural number n with $k^2 < n < (k + 1)^2$ for some $k \in \mathbb{N}$ is the square of a rational; take $(n, k) = (7, 2)$, as $4 < 7 < 9$, and $(n, k) = (14, 3)$, as $9 < 14 < 16$. So 7 has no square root in $\mathbb{Q}(\sqrt{2})$. This is the method of *Number Systems: From* $\mathbb{N}$ *to* $\mathbb{R}$, Chapter 6, which showed the same for 3.

Exercise. R3

Show that no relation $<$ makes $\mathbb{Q}(i)$ (Example (Rational and Integer Complex Numbers)) an ordered field, and none makes the four-element field F of Exercise C1 of Section 2.2 an ordered field.

Solution. F. In F, $1 + 1 = (1 + 1) + (0 + 0)\alpha = 0$, by the definition of the sum, with coefficients in $\mathbb{F}_2$. So $2 \cdot 1 = 0$ and $\operatorname{char} F \neq 0$, while every ordered field has characteristic 0 (Example (The characteristics of the standard fields)). This is the method of Theorem 1.3.3 (e) and Number Systems: From \(\nN\) to \(\nR\), Example (A field with two elements that cannot be ordered): in an ordered field $n \cdot 1 > 0$ for $n \geq 1$ (Number Systems: From \(\nN\) to \(\nR\), Proposition 2.4.3 (h)), while here $2 \cdot 1 = 0$.

$\mathbb{Q}(i)$. This is the method of Analytic Preliminaries, Proposition 1.4.3; the proposition itself does not apply, since an order on the subfield $\mathbb{Q}(i)$ need not extend to $\mathbb{C}$. Suppose $<$ made $\mathbb{Q}(i)$ an ordered field. Then $i \neq 0$ gives $i^2 > 0$, and $1 \neq 0$ gives $1 > 0$ (Number Systems: From \(\nN\) to \(\nR\), Proposition 2.4.3 (e)). But $i^2 = -1$ (Analytic Preliminaries, Proposition 1.1.1; the operations of $\mathbb{Q}(i)$ are those of $\mathbb{C}$), and $1 > 0$ gives $-1 < 0$ (Number Systems: From \(\nN\) to \(\nR\), Proposition 2.4.3 (a)). So $0 < -1 < 0$, and $0 < 0$ by transitivity, which is impossible for the strict part of an order. The characteristic does not help here, since $\operatorname{char} \mathbb{Q}(i) = 0$ (Example (The characteristics of the standard fields)): the obstruction for $\mathbb{Q}(i)$ is a square, not a multiple of 1.

Exercise. R4

Find all homomorphisms $\mathbb{Q}(\sqrt{2}) \to \mathbb{R}$.

Solution. There are exactly two: the inclusion $a + b\sqrt{2} \mapsto a + b\sqrt{2}$, and $a + b\sqrt{2} \mapsto a - b\sqrt{2}$.

Every homomorphism is one of them. Let $\sigma\colon \mathbb{Q}(\sqrt{2}) \to \mathbb{R}$ be a homomorphism. The inclusions $\mathbb{Q} \to \mathbb{Q}(\sqrt{2})$ and $\mathbb{Q} \to \mathbb{R}$ are homomorphisms (Example (Conjugation and Other Automorphisms) (ii); $\mathbb{Q}$ is a subfield of both), so they are $\iota_{\mathbb{Q}(\sqrt{2})}$ and $\iota_{\mathbb{R}}$ by the uniqueness clause of Theorem 2.5.5 (b) (both fields have characteristic 0, Example (The characteristics of the standard fields)). By Corollary 2.5.6 (a), $\sigma \circ \iota_{\mathbb{Q}(\sqrt{2})} = \iota_{\mathbb{R}}$: $\sigma(q) = q$ for every $q \in \mathbb{Q}$. Then

$$\sigma(\sqrt{2})^2 = \sigma((\sqrt{2})^2) = \sigma(2) = 2 = (\sqrt{2})^2,$$

so $\sigma(\sqrt{2}) = \sqrt{2}$ or $\sigma(\sqrt{2}) = -\sqrt{2}$ by Corollary 2.1.2 in $\mathbb{R}$. Hence $\sigma(a + b\sqrt{2}) = \sigma(a) + \sigma(b)\sigma(\sqrt{2}) = a \pm b\sqrt{2}$.

Both occur. The inclusion is a homomorphism (Example (Conjugation and Other Automorphisms) (ii)), and so is conjugation $a + b\sqrt{2} \mapsto a - b\sqrt{2}$ of $\mathbb{Q}(\sqrt{2})$ followed by the inclusion (Example (Conjugation and Other Automorphisms) (iii) and Proposition 2.4.1 (e)). They differ at $\sqrt{2}$, since $\sqrt{2} > 0 > -\sqrt{2}$.

The tool, and the look-alikes. The tool is rigidity on the prime subfield (Corollary 2.5.6) followed by "push the equation $x^2 = 2$ through σ". The uniqueness theorem for complete ordered fields of *Number Systems: From* $\mathbb{N}$ *to* $\mathbb{R}$, Chapter 6 (Number Systems: From \(\nN\) to \(\nR\), Theorem 6.3.4) does not apply: it concerns maps of a complete ordered field to itself, and $\mathbb{Q}(\sqrt{2})$ is not complete. Nor does "a homomorphism into an ordered field preserves order": the second homomorphism sends $\sqrt{2} > 0$ to $-\sqrt{2} < 0$.